

Luc's Statistical Coffee Price Forecast

Monthly Model Report

2026-09-01

Executive Summary

Coffee Price Forecast Summary

Our forecast calls for Arabica prices to decline gradually over the next three months, starting at 342.3 *¢/lb* in month one and settling near 323.3 *¢/lb* by month three. Robusta is expected to trade sideways in a narrow range around 182 *¢/lb*, with minimal directional movement. Uncertainty widens as we look further ahead: the 80% forecast range for Arabica spans about 60-64 *¢/lb*, while Robusta's tighter range spans roughly 31 *¢/lb*.

The model has demonstrated solid near-term accuracy. One-month forecasts carry a 6-6% average forecast error for both varieties, with our 80% confidence bands capturing actual prices at acceptable rates versus an 80% target. Accuracy declines as the forecast horizon extends, with three-month errors reaching 11-12%, a typical pattern for commodity forecasting.

The spread signal between Arabica and Robusta sits at a neutral reading with a half-life of 23 months. This suggests the current price relationship is neither stretched nor compressed relative to its long-term average, pointing to stable relative value between the two coffee grades over the forecast period.

Current Forecasts

Median (p50) price forecasts and 80% confidence intervals (p10–p90) for the next three months.

Symbol	Horizon	Median (¢/lb)	80% CI (¢/lb)	Band width
Arabica (KC=F)	h=1	342.3	[317.2, 380.8]	63.6
Arabica (KC=F)	h=2	330.8	[306.5, 368.0]	61.4
Arabica (KC=F)	h=3	323.3	[299.6, 359.6]	60
Robusta (RM=F)	h=1	182.1	[167.5, 198.5]	30.9
Robusta (RM=F)	h=2	183.8	[169.0, 200.3]	31.2
Robusta (RM=F)	h=3	181.5	[166.9, 197.7]	30.8

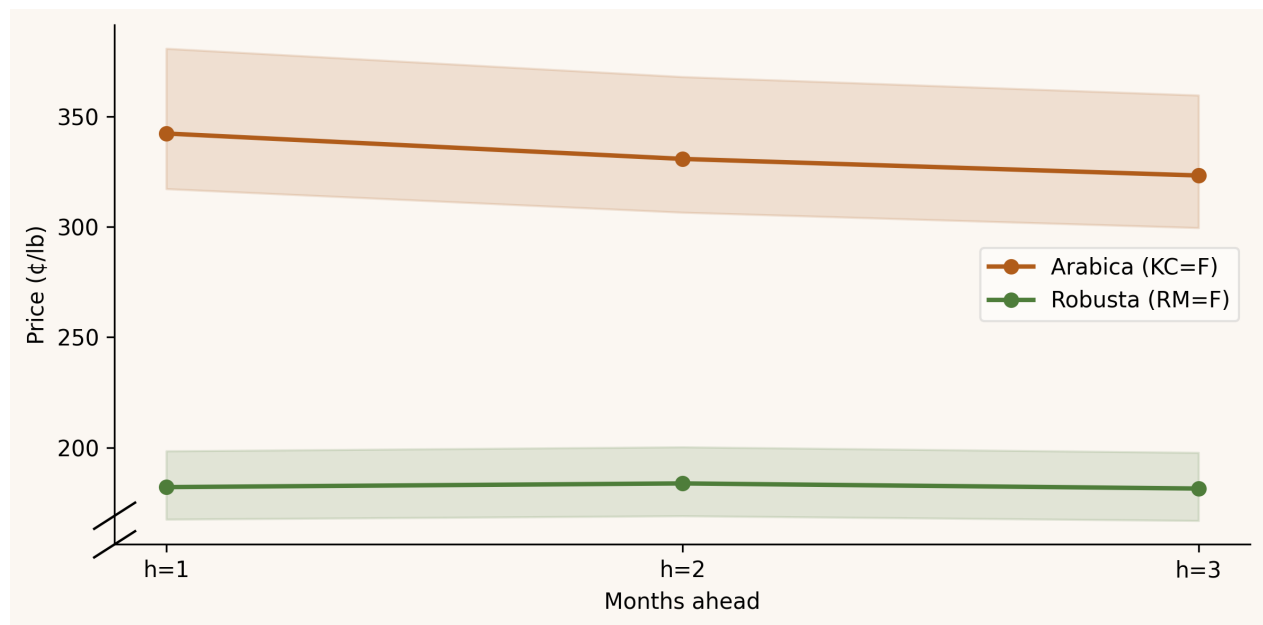


Figure 1: Price trajectory with 80% confidence ribbon

Model Performance

Mean absolute error (MAE), mean absolute percentage error (MAPE), and 80% interval coverage across all walk-forward backtest windows.

Symbol	Horizon	MAE (¢/lb)	MAPE	Coverage 80%
Arabica	h=1	12.8	6.1%	N/A
Arabica	h=2	19.8	9.2%	N/A
Arabica	h=3	25.3	11.4%	N/A
Robusta	h=1	7	5.6%	N/A
Robusta	h=2	11.3	9.1%	N/A
Robusta	h=3	15.1	11.9%	N/A

Coverage 80% target: 80% of forecast ranges should contain the realized price. Values below 80% indicate the model is over-confident.

Recent Forecast Performance

One-month-ahead forecasts versus realized prices over the last three months.

Month	Symbol	Forecast (¢/lb)	Actual (¢/lb)	Error
2026-03	Arabica (KC=F)	312.2	334.1	-6.6% (below actual)
2026-03	Robusta (RM=F)	170.9	176.8	-3.3% (below actual)
2026-02	Arabica (KC=F)	360.8	321.4	+12.3% (above actual)
2026-02	Robusta (RM=F)	192.2	179.6	+7.0% (above actual)
2026-01	Arabica (KC=F)	371.1	363.9	+2.0% (above actual)
2026-01	Robusta (RM=F)	182.2	192.3	-5.2% (below actual)

January 2026

The model forecast Arabica at 371.1 ¢/lb and Robusta at 182.2 ¢/lb, coming within 2.0% for Arabica but missing Robusta by 5.2% on the low side as prices reached 192.3 ¢/lb. Robusta's outperformance likely reflected supply tightness in Vietnam and Indonesia that historical correlations underweighted, while currency weakness in major producing regions may have supported prices beyond what the model's backward-looking framework anticipated.

February 2026

Both contracts saw significant overestimation, with Arabica forecast at 360.8 ¢/lb versus an actual 321.4 ¢/lb (12.3% miss) and Robusta at 192.2 ¢/lb against 179.6 ¢/lb (7.0% miss). The sharp pullback suggests the model failed to anticipate a demand shock or a shift in risk sentiment, possibly from unexpected macroeconomic signals or profit-taking after the January rally that statistical models calibrated on longer-term relationships typically struggle to capture.

March 2026

The model underestimated both contracts, forecasting Arabica at 312.2 ¢/lb versus 334.1 ¢/lb (6.6% low) and Robusta at 170.9 ¢/lb versus 176.8 ¢/lb (3.3% low). The recovery from February lows outpaced the model's expectations, likely driven by weather concerns in key growing regions or fund repositioning that wasn't embedded in the historical price structure the model relied on.

Arabica–Robusta Spread Signal

The spread model tracks whether Arabica is unusually cheap or expensive relative to Robusta. It measures the **z-score** of the log price ratio — how many standard deviations the current ratio sits from its long-run average. A z-score near zero means the relationship is normal; a z-score beyond ± 2 means a rare dislocation (this happens roughly 5% of the time historically) that has tended to revert. A trade signal fires at $|z| > 2$ and closes once the ratio normalises back to $|z| < 0.5$.

As of	2026-07-01
Current signal	Neutral — ratio within normal range, no edge detected
Z-score	0.18 std devs from average
Half-life	22.7 months — typical time to revert to average
Entry threshold	
Exit threshold	

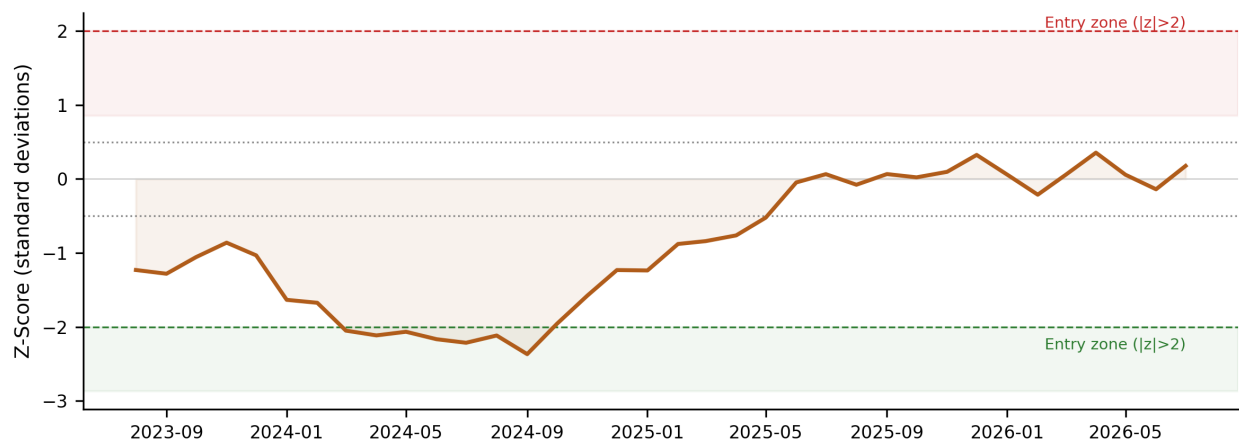


Figure 2: Z-score of Arabica/Robusta log spread — last 36 months. Red zone = Arabica expensive; green zone = Arabica cheap. Dashed lines mark entry thresholds (± 2 std devs); dotted lines mark exit thresholds (± 0.5).

Data & Methodology

Data sources: Arabica (KC=F) and Robusta (RM=F) monthly prices from FRED (PCOFFOTMUSDM, PCOFFROBUSDM). FX drivers (BRL/USD, VND/USD, IDR/USD) from Alpha Vantage. DXY broad index from FRED (DTWEXBGS). Data starts 2000-01-01; monthly series is the mean of daily adjusted closes within each calendar month.

VECM: A two-variable Vector Error Correction Model is fit on log Arabica and log Robusta prices, with BRL, VND, IDR, and DXY as exogenous drivers. Cointegration rank is fixed at 1 (confirmed by Engle-Granger test, $p=0.009$). Lag order is selected by AIC each run. Point forecasts at 1, 2, and 3 months use a naïve FX assumption (hold FX at last observed value).

GAMLSS SHASH: A Sinh-ArcSinh distribution is fit (via R's `gamlss` package) to the VECM forecast residuals, separately per symbol and per volatility regime (Low / Medium / High, defined by rolling 12-month standard deviation). This gives a full predictive distribution. Final price quantiles are `price_qX = vecm_point * exp(gamlss_residual_qX)`.

Spread signal: An AR(1) model is fit to the log spread series; the expanding z-score triggers entry at $|z| > 2$ and exit at $|z| < 0.5$.

Report generated by the `coffee-forecast` pipeline.